



VIRGINIA TECH[®]

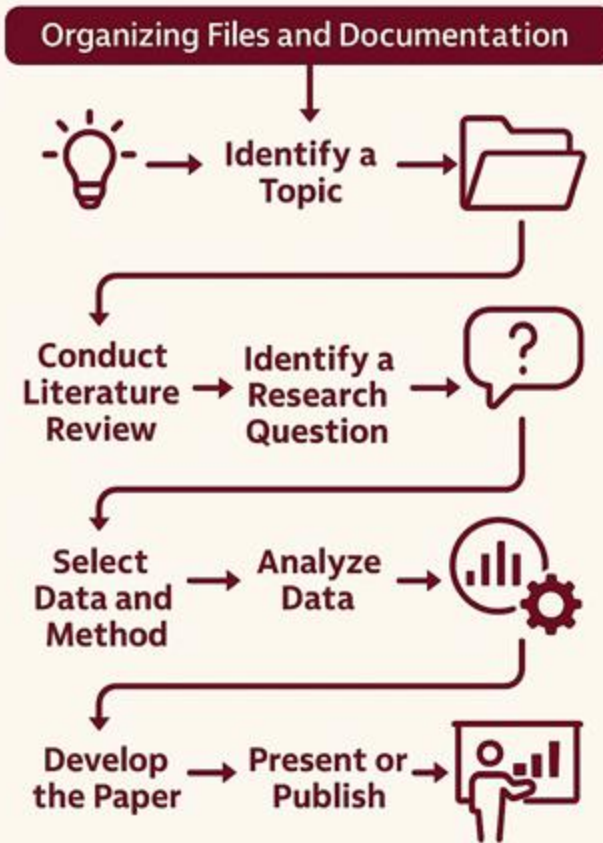


A Practical Guide to *Doing* Quantitative Social Science Research

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DSPG Summer Program

The Empirical Research Process

Mapping the Research Process



Workshop Goals

- I. Identify an Empirical Research Question:
 - A Question
 - A Meaningful Question
 - A Meaningful, Feasible Question
- I. Implementation Details
 - Refine Your Question
 - Understand and Contribute to Literature
 - Match Data Analytical Tools to Questions
- I. Common Setbacks in Economics Research
- II. Maintain Organization

I. Empirical Research Question

A Question

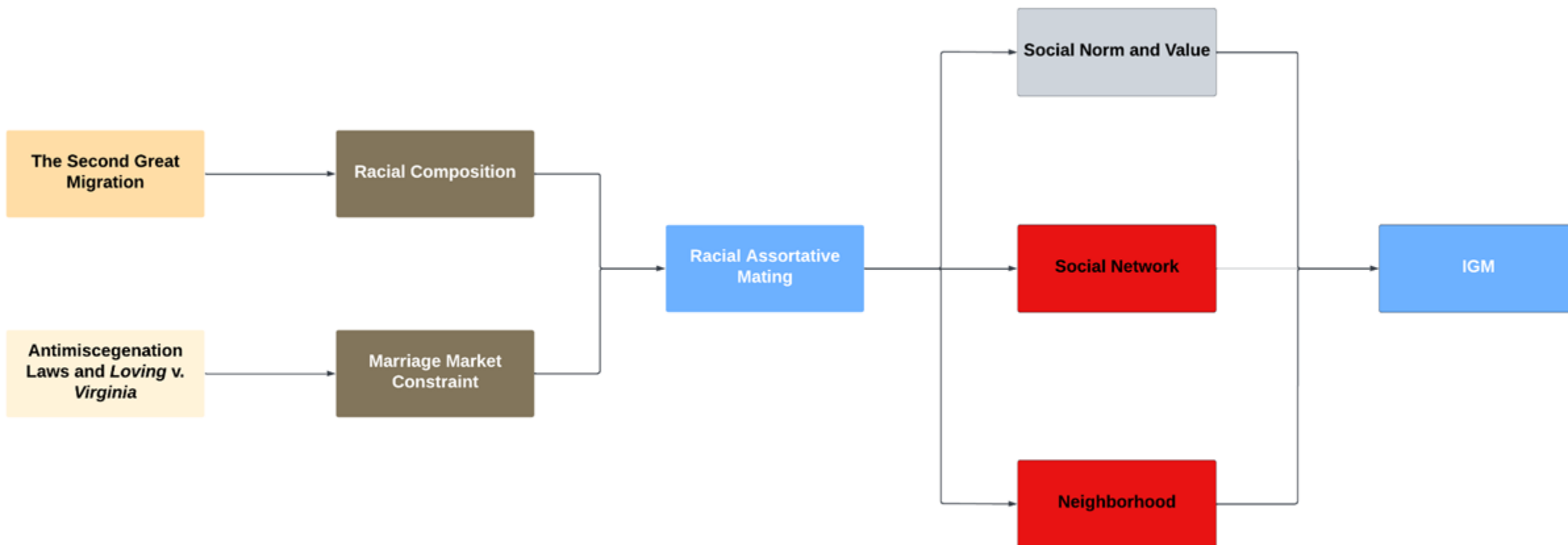
What is a research question?

I. Simplistic form (whether)

$A \rightarrow B$

I. More sophisticated form (why)

$A \rightarrow C \rightarrow B$ (Economically and statistically)



What is a research question?

Understanding the question with an (economic) framework

Example: Crop project

1. Expected export and actual yield (gap): (increase price)
 2. Ending stock (lower price)
-
- a. Not an economic overarching framework
 - b. Not direct information (their ultimate goal is price, multiple inputs to predict better)

What is a research question?

Understanding the question with an (economic) framework

Example: childcare cost and female labor force participation

1. Negative?
2. Women work at childcare to reduce the cost! (Grace)
3. People respond to offset the cost by substituting (Chunbei and I both work, but in academia with flexible schedules and remote work)

What is a research question?

Layers of good research:

1. New B, and New A
2. New A, or New B
3. New ways to answer old $A \rightarrow B$

I. Empirical Research Question

A Meaningful Question

What Makes a Meaningful Research Question?

Speaks to a real-world issue, policy, or economic behavior

- Informs decision-making
 - Policy: would a government agency, nonprofit, or business care about the answer?
 - Stakeholder: Can the findings affect how resources are allocated, or rules are made?
 - Individual: Does this information help with individual decisions?
- Which is better?
 - How often do farmers attend farmers markets?
 - How does regular participation in farmers markets affect small farm income?

What Makes a Meaningful Research Question?

Builds on, extends, or challenges existing literature

- a. A new question, data, method, event
- b. An improvement

I. Empirical Research Question

A Meaningful, Feasible Question

What Makes a Feasible Research Question?

Answerable with available data and focused

- Which of the following is a better research question?
 - Is immigration good or bad?
 - What is the effect of undocumented immigration on low-skilled natives' wages?
- Tip: clearly defined A and B that can be measured with data.
- Example: Childcare group found great datasets, but it is challenging to have access to.

What Makes a Feasible Research Question?

Able to isolate a causal relationship and its mechanisms

- Example
 - Does legal status improve the mental health outcomes of undocumented immigrants?
 - How?
- Which is a more feasible research question?
 - Does poverty cause low educational attainment?
 - Does free school lunch program improve high school graduation rates?

At the end

We advance cutting-edge data analytics by developing and applying novel methods for analyzing complex datasets. Our research enables more robust and rigorous empirical research, helping researchers and practitioners solve real-world problems, and generating insights that drive impactful decision-making.

2024: Ongoing research

Beyond common support: an iterative approach to nonseparable models with endogeneity

Authors +

What we did +

Why we did it +

The impact +

Rehabilitating the Once-Abandoned Endogenous IV

Authors +

What we did +

Why we did it +

The impact +

Kohl Centre Format!

II. Implementation Details

Refine Your Question

From Topic to Research Question

- Activity:
 - Topic – Minimum Wage
 - How can we turn this into a focused, empirical question?
- Your DSPG project:
 - What is your project's research question?
 - Is it a strong research question?

II. Implementation Details

Understand and Contribute
to Literature

Understanding the Literature

- Skim abstracts, intros, conclusions first
- Track:
 - Research question (X and Y)
 - Data and method
 - Key findings
- Build a literature matrix

Organizing the Literature

Author(s)	Year	Journal	Title	Research Question	Data - specific source for all variables	Method	Key Findings	Notes

Contribute to the Literature

- What is a contribution?
 - New question or under-explored topic
 - New X variable
 - New Y variable
 - New geographic area: studied before in another country, but not your region.
 - New data: uses recent data or better data
 - New method: newly developed method
 - New event: a policy or event that has not been utilized

Activity: What is the contribution?

- Title: Do Compulsory Schooling Laws Affect Fertility Behaviors and Marriages? Evidence from India
- Abstract: This paper exploits an education policy in India generated by a 2010 schooling reform to examine the effect of education on women's family planning decisions. The key element of the reform was that it required students to complete eight years of primary education (age 6-14 years). I employ an instrumental variable difference-in-difference approach that measures the exogenous variation in treatment intensity in different states across birth cohorts of women measured by birth year and birth months. The reform led to an increase in total years of education, a delay in the age at first marriage, a postponement of sexual activity, and reduced fertility beginning at the age of 22. I also examine the potential mechanisms through which increased education affects fertility. The findings suggest early use of modern contraceptives, reduction in the marital education gap, increased literacy, and utilization of healthcare services contribute to reduced fertility. These results are consistent with the increased empowerment of women influencing their fertility decision thereby highlighting the importance of the government's efforts to promote education through policy initiatives.

II. Implementation Details

Match Data Analytic tools
to Questions

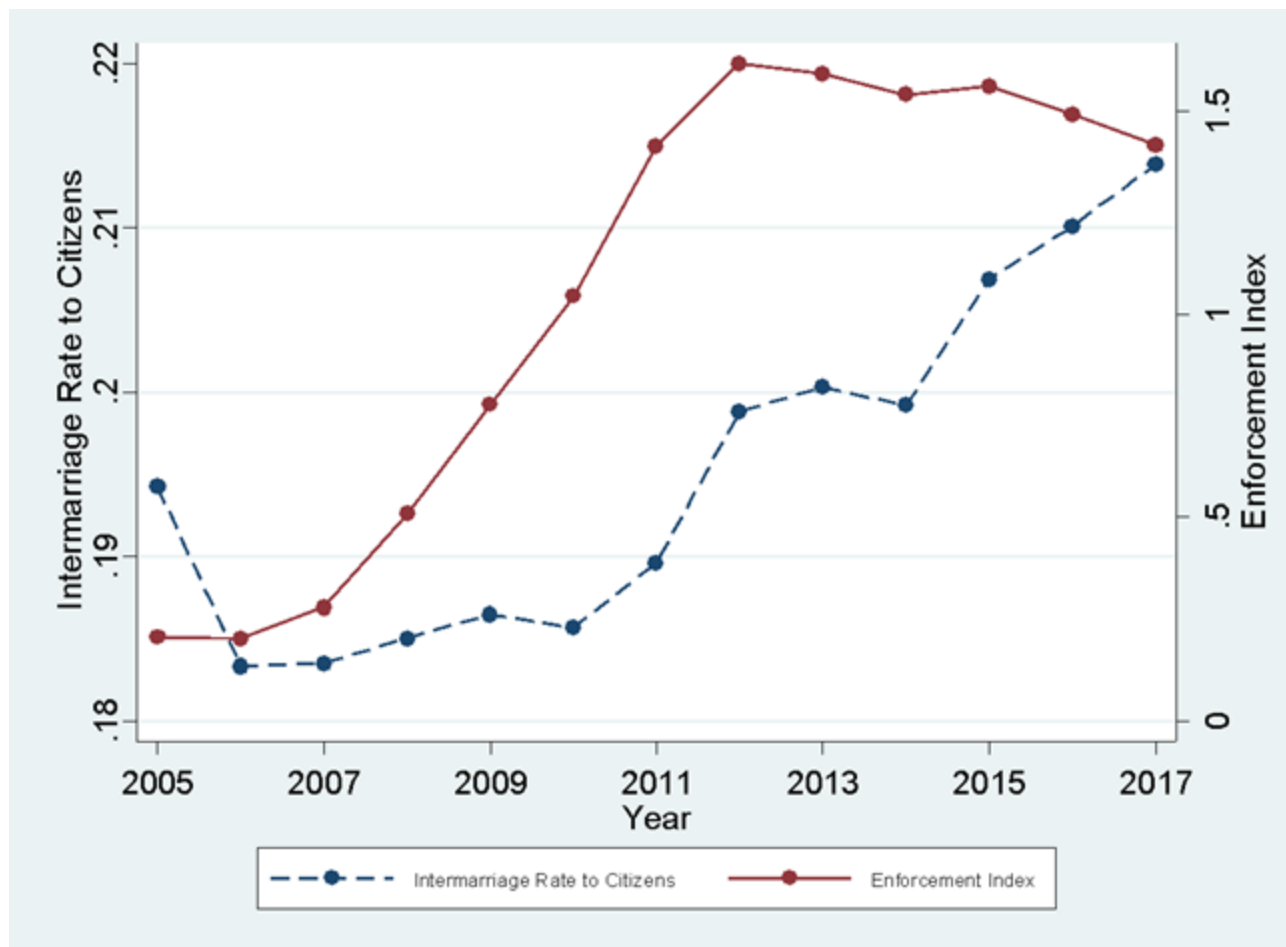
Matching Methods to Questions

- Descriptive: What's happening?
- Predictive: What will happen?
- Causal: Does X cause Y?
- Common Empirical Methods
 - OLS regression, fixed effects
 - Difference-in-differences, Randomized Control Experiments, Instrument variables, Regression discontinuity, Event studies, Time series methods
 - Structural models, machine learning

Expectations for Rigor

- Clear identification strategy
- Understand assumptions in empirical methods
- Robustness checks
- Transparency – replicable code and data

Activity: Can this graph prove causality?



- What more do you need to prove X causes Y?

III. Common Setbacks in Economics Research

What could go wrong?

- Messy or missing data
 - Incomplete entries, inconsistent formats, coding errors, unreasonable values, sample size too small
- Strategy: have a data cleaning plan
 - Read raw data documentation carefully
 - Understand variable values and coding
 - Preserve raw files
 - Adopt common practice in literature
 - Explore alternative data sources
 - Document assumptions

What could go wrong?

- Unexpected or “wrong” results
 - Findings contradict expectations or literature
- Strategy:
 - Re-check data and code: Are you using the right sample, units, variables?
 - Run descriptive statistics: Are there outliers, odd distributions, unexpected trends?
 - Try robustness checks: Change the model specification – add/remove controls
 - Try different subsamples: Are there heterogeneity
 - Return to the literature and theory: Could your context be different?
 - Reframe your research question or contribution.
- The most interesting papers often begin with: *We found something we didn't expect... and we figured out why.*

What could go wrong?

- Unable to replicate results
 - Minor changes lead to big differences
 - Hard to replicate
- Strategy:
 - Modularize your code
 - Break into clear, logical chunks
 - Helps isolate where changes are affecting results
 - Fix your random seed if randomness is involved
 - Clear documentation: summarize file purposes and key decisions.
 - Create a replication script
 - One script that runs the full analysis from start to finish.
 - Use version control

What could go wrong?

- Disconnected writing and analysis
 - Results don't clearly answer the question posed
- Strategy:
 - Keep the research question in front of you – literally
 - Ask: does this result directly help answer the question.
 - Draft abstract and intro early, revise often
 - Reverse engineer your tables and figures
 - Look at your output and ask: What question does this answer?

IV. Maintain Organization

Tools and Documentation

- Version control
 - Git or manual
- README files
- Research logs
 - Problems encountered
 - Decisions made
- Code commenting

Final Takeaways

- Clarity, contribution, method, rigor
- Keep your work replicable and your process documented
- Ask questions early and often!